

ISRO 2025: Topic-Wise Practice Questions

TOPIC 3: COMPUTER NETWORKS

Based on 2023 ISRO Questions: 12, 13, 14, 15, 16

****Coverage Analysis****: This topic had 5 questions (6.25%) in 2023 paper, but with heavy emphasis on numerical calculations and protocol knowledge.

****Key Concepts Tested****: Network performance calculations, protocol layer overhead, wireless standards, error detection/correction codes, frame size calculations.

NETWORK PERFORMANCE & CALCULATIONS (Questions 1-15)

****1. A 2 Gbps link with 30ms RTT uses 1500-byte packets. For >98% channel utilization, minimum window size should be:****

- (a) 3000 packets
- (b) 5000 packets
- (c) 6000 packets
- (d) 7500 packets

****2. Bandwidth-delay product represents:****

- (a) Maximum data in transit
- (b) Network throughput
- (c) Propagation delay
- (d) Transmission delay

****3. For satellite communication with 270ms RTT and 1 Mbps bandwidth, bandwidth-delay product is:****

- (a) 270 KB
- (b) 33.75 KB
- (c) 135 KB
- (d) 540 KB

****4. Utilization efficiency formula for Stop-and-Wait protocol:****

- (a) $1/(1+2a)$ where $a = \text{Propagation delay/Transmission delay}$
- (b) $1/(1+a)$
- (c) $1/(2+a)$
- (d) $a/(1+a)$

****5. Go-Back-N ARQ with window size W and error probability p has efficiency:****

- (a) $W(1-p)/(1+2ap)$
- (b) $(1-p)/(1+2ap)$
- (c) $W(1-p)$
- (d) $(1-p)W/(1+a)$

****6. Selective Repeat ARQ efficiency is:****

- (a) Same as Go-Back-N
- (b) Better than Go-Back-N
- (c) Worse than Go-Back-N
- (d) Independent of error probability

****7. Propagation delay for 1000 km fiber optic cable (speed = 2×10^8 m/s):****

- (a) 5 ms
- (b) 10 ms
- (c) 15 ms
- (d) 20 ms

****8. Transmission delay for 1MB file on 10 Mbps link:****

- (a) 0.8 seconds
- (b) 1.6 seconds
- (c) 0.4 seconds
- (d) 2 seconds

****9. Total delay = Transmission + Propagation + Processing + Queuing. For real-time applications, most critical is:****

- (a) Transmission delay
- (b) Propagation delay
- (c) Processing delay
- (d) Queuing delay

****10. Network with 5 routers in series, each adding 2ms processing delay. Total processing delay:****

- (a) 2 ms
- (b) 5 ms
- (c) 10 ms
- (d) 15 ms

****11. Frame size optimization depends on:****

- (a) Error rate
- (b) Overhead
- (c) Retransmission cost
- (d) All of above

****12. Optimal frame size minimizes:****

- (a) Transmission time
- (b) Error probability
- (c) Total transmission time including retransmissions
- (d) Overhead

****13. Little's Law: Average number in system = Average arrival rate $\times$ Average time in system. If arrival rate = 100 packets/sec, average delay = 50ms:****

- (a) 5 packets
- (b) 50 packets
- (c) 500 packets

(d) 2000 packets

****14. Throughput vs Bandwidth:****

- (a) Throughput $\leq$ Bandwidth always
- (b) Throughput $>$ Bandwidth possible
- (c) Throughput = Bandwidth always
- (d) No relationship

****15. For CSMA/CD, minimum frame size ensures:****

- (a) Collision detection before transmission ends
- (b) Maximum utilization
- (c) Error correction
- (d) Flow control

**PROTOCOL LAYERS & OVERHEAD (Questions 16-25)**

****16. 7-layer protocol with applications generating 500-byte messages, each layer adds 10-byte header. Bandwidth waste percentage:****

- (a) 12%
- (b) 14%
- (c) 16%
- (d) 18%

****17. TCP header size without options:****

- (a) 20 bytes
- (b) 24 bytes
- (c) 32 bytes
- (d) 40 bytes

****18. IP header minimum size:****

- (a) 20 bytes
- (b) 24 bytes
- (c) 32 bytes
- (d) 40 bytes

****19. Ethernet frame header size:****

- (a) 14 bytes
- (b) 18 bytes
- (c) 22 bytes
- (d) 26 bytes

****20. Maximum Transmission Unit (MTU) for Ethernet:****

- (a) 1024 bytes
- (b) 1500 bytes
- (c) 2048 bytes
- (d) 4096 bytes

****21. Protocol overhead efficiency = Data/(Data+Headers). For 1000-byte data with 60-byte total headers:****

- (a) 94.3%
- (b) 96.2%
- (c) 92.5%
- (d) 98.1%

****22. IPv6 header size:****

- (a) 20 bytes
- (b) 40 bytes
- (c) 60 bytes
- (d) Variable

****23. UDP header size:****

- (a) 4 bytes
- (b) 8 bytes
- (c) 12 bytes
- (d) 16 bytes

****24. Frame overhead includes:****

- (a) Headers only
- (b) Trailers only
- (c) Headers and trailers
- (d) Data only

****25. Encapsulation adds headers at:****

- (a) Source only
- (b) Destination only
- (c) Each layer during transmission
- (d) Routers only

**WIRELESS NETWORKS & STANDARDS (Questions 26-35)**

****26. Match 802.11 versions with speeds:****

- (i) 802.11b (P) 54 Mbps
- (ii) 802.11g (Q) 11 Mbps
- (iii) 802.11n (R) 600 Mbps
- (iv) 802.11ac (S) 3.5 Gbps

- (a) (i)-(Q), (ii)-(P), (iii)-(R), (iv)-(S)
- (b) (i)-(P), (ii)-(Q), (iii)-(R), (iv)-(S)
- (c) (i)-(Q), (ii)-(P), (iii)-(S), (iv)-(R)
- (d) (i)-(P), (ii)-(R), (iii)-(Q), (iv)-(S)

****27. 802.11 uses which access method:****

- (a) CSMA/CD
- (b) CSMA/CA
- (c) TDMA

(d) FDMA

****28. Hidden terminal problem in wireless networks is solved by:****

- (a) RTS/CTS
- (b) Acknowledgments
- (c) Increased power
- (d) Better antennas

****29. Exposed terminal problem occurs when:****

- (a) Terminal cannot hear other transmissions
- (b) Terminal hears transmissions not intended for it
- (c) Terminal is out of range
- (d) Terminal has low battery

****30. Wi-Fi operates in which frequency bands:****

- (a) 2.4 GHz only
- (b) 5 GHz only
- (c) 2.4 GHz and 5 GHz
- (d) 6 GHz only

****31. Bluetooth uses:****

- (a) Frequency hopping spread spectrum
- (b) Direct sequence spread spectrum
- (c) Time division multiple access
- (d) Code division multiple access

****32. WPA2 uses which encryption:****

- (a) WEP
- (b) AES
- (c) DES
- (d) 3DES

****33. 802.11ax (Wi-Fi 6) improvement over 802.11ac:****

- (a) Higher speed only
- (b) Better efficiency only
- (c) Both higher speed and better efficiency
- (d) Lower power consumption only

****34. Cellular network generations:****

1G - Analog, 2G - Digital, 3G - Data, 4G - ?, 5G - ?

- (a) 4G - Broadband, 5G - Ultra-fast
- (b) 4G - LTE, 5G - mmWave
- (c) 4G - All IP, 5G - IoT
- (d) 4G - Mobile broadband, 5G - Ultra-low latency

****35. MIMO technology provides:****

- (a) Multiple antennas for better reception
- (b) Spatial diversity

- (c) Increased data rates
- (d) All of above

ERROR DETECTION & CORRECTION (Questions 36-45)

36. Which is true about error detection/correction:

- (a) Parity and CRC are both error correcting
- (b) Parity and Hamming are both error correcting
- (c) Reed-Solomon and LDPC are both error correcting
- (d) CRC and LDPC are both error correcting

37. Single parity bit can:

- (a) Detect single bit errors only
- (b) Correct single bit errors
- (c) Detect double bit errors
- (d) Correct double bit errors

38. Hamming code with 4 data bits requires how many parity bits:

- (a) 2
- (b) 3
- (c) 4
- (d) 5

39. CRC polynomial x^3+x+1 can detect:

- (a) Single bit errors
- (b) Double bit errors
- (c) Burst errors up to 3 bits
- (d) All of above

40. Checksum method:

- (a) Adds all data words
- (b) Takes complement of sum
- (c) Sends complement as checksum
- (d) All of above

41. Reed-Solomon codes are used in:

- (a) CD/DVD error correction
- (b) Satellite communication
- (c) Storage systems
- (d) All of above

42. LDPC (Low Density Parity Check) codes are:

- (a) Block codes
- (b) Convolutional codes
- (c) Linear codes
- (d) Both (a) and (c)

43. Error correction capability of Hamming(7,4) code:

- (a) Detect 1, correct 0 errors
- (b) Detect 2, correct 1 error
- (c) Detect 3, correct 2 errors
- (d) Detect 4, correct 3 errors

****44. Interleaving is used to:****

- (a) Spread burst errors
- (b) Reduce random errors
- (c) Improve correction capability
- (d) Both (a) and (c)

****45. Forward Error Correction (FEC) vs Automatic Repeat Request (ARQ):****

- (a) FEC corrects errors, ARQ detects and retransmits
- (b) FEC is faster for real-time applications
- (c) ARQ is better for reliable data transfer
- (d) All of above

**NETWORK PROTOCOLS & STANDARDS (Questions 46-55)**

****46. OSI model layer responsible for routing:****

- (a) Physical
- (b) Data Link
- (c) Network
- (d) Transport

****47. TCP provides:****

- (a) Reliable delivery
- (b) Flow control
- (c) Congestion control
- (d) All of above

****48. UDP is preferred over TCP for:****

- (a) File transfer
- (b) Email
- (c) Real-time applications
- (d) Web browsing

****49. HTTP status code 404 means:****

- (a) Server error
- (b) Client error - not found
- (c) Redirection
- (d) Success

****50. HTTPS uses which port by default:****

- (a) 80
- (b) 443
- (c) 8080
- (d) 21

****51. DNS uses which transport protocol:****

- (a) TCP only
- (b) UDP only
- (c) Both TCP and UDP
- (d) ICMP

****52. ARP (Address Resolution Protocol) maps:****

- (a) IP to MAC address
- (b) MAC to IP address
- (c) Domain name to IP
- (d) Port number to service

****53. ICMP is used for:****

- (a) Error reporting
- (b) Network diagnostics
- (c) Control messages
- (d) All of above

****54. NAT (Network Address Translation) provides:****

- (a) Security through address hiding
- (b) IPv4 address conservation
- (c) Multiple devices sharing single public IP
- (d) All of above

****55. Quality of Service (QoS) mechanisms include:****

- (a) Traffic shaping
- (b) Priority queuing
- (c) Bandwidth reservation
- (d) All of above

**ANSWER KEY - COMPUTER NETWORKS**

****1-10:**** (b), (a), (b), (a), (b), (b), (a), (a), (d), (c)

****11-20:**** (d), (c), (a), (a), (a), (a), (a), (a), (a), (b)

****21-30:**** (a), (b), (b), (c), (c), (a), (b), (a), (b), (c)

****31-40:**** (a), (b), (c), (d), (d), (c), (b), (d), (d), (d)

****41-50:**** (d), (d), (b), (d), (d), (c), (d), (c), (b), (b)

****51-55:**** (c), (a), (d), (d), (d)

**Critical Study Focus for September 14th:**

1. ****Network Calculations (Q1-15)**:** Master bandwidth-delay product, utilization efficiency, frame size calculations

2. **Protocol Overhead (Q16-25)**: Understand how headers affect efficiency
3. **Wireless Standards (Q26-35)**: Memorize 802.11 speeds and characteristics
4. **Error Control (Q36-45)**: Focus on detection vs correction capabilities

Important Formulas to Remember:

- Utilization = $1/(1+2a)$ where a = Propagation delay/Transmission delay
- Bandwidth-delay product = Bandwidth $\times$ Round trip time
- Overhead % = $(\text{Headers}/(\text{Data}+\text{Headers})) \times 100$
- Total delay = Transmission + Propagation + Processing + Queuing

Time Allocation: Spend 25% of network study time on numerical problems as ISRO heavily tests calculations.